

ME-802 (GS)
B.E. VIII Semester
Examination, May 2019
Grading System (GS)
Machine Design
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the advantages and disadvantages of flat belt drive?
b) Discuss the construction of wire ropes with the help of a neat sketch.
2. a) What is the use of Lagrange multipliers? What is their practical significance?
b) Find the maxima and minima, if any of the functions
 - i) $f(x) = 4x^3 - 18x^2 + 27x - 7$
 - ii) $f(x) = x^4 / (x-1)(x-3)^3$
3. Design a chain drive to actuate a compressor from 15 kW electric motor running at 1000 rpm, the compressor speed being 350 rpm. The minimum centre distance is 500 mm. The compressor operates 16 hours per day. The chain tension may be adjusted by shifting the motor on slides.

[2]

4. A pair of parallel helical gears consists of an 18 teeth pinion meshing with a 45 teeth gear. 7.5 kW power at 2000 rpm is applied to the pinion through its shaft. The normal module is 6 mm while the normal pressure angle is 20° . The helix angle is 23° . Determine the tangential, radial and axial components of the resultant tooth force between the meshing teeth. Draw also free body diagram of forces. <https://www.rgpvonline.com>
5. A crank shaft of the center-crank type is to be designed for a diesel engine developing 7.5 kW on brake at 1200 r.p.m. The crank shaft is of the forged type and has to carry two y wheels of 500 N each on either side of the main bearing. The plane of rotation of each wheel is 100mm from the center line of adjacent main bearing. The maximum load on connecting rod is 35kN. The length of stroke of piston is 160mm and the length of connecting rod 320mm. Maximum torque will be experienced in the crank shaft when the crank turns 300° from the inner dead center position. The maximum permissible stress in crank pin, web and shaft should not exceed 60 N/mm^2 , the safe limit for the bearing stress for the crank pin and the main bearings is 800 N/mm^2 . Design the crank shaft.
6. A V-belt is driven on a flat pulley and a V-pulley. The drive transmits 20 kW from a 250mm diameter V-pulley operating at 1800 rpm to a 900mm diameter flat pulley. The centre distance is 1m, the angle of groove 40° and $\mu = 0.2$. If density of belting is 1110 kg/m^3 and allowable stress is 2.1 MPa for belt material, what will be the number of belts required if C-size V-belts having 230 mm^2 cross-sectional area are used.
7. a) What do you mean by slip in a belt drive? Differentiate between open belt and crossed belt drive. List out the applications of belt drives.

b) Following are the details of a crossed belt drive

Diameter of the driver	:	200mm
Diameter of the follower	:	400mm
Center distance of the drive	:	2m
Speed of the drive	:	400rpm
Angle of contact	:	197.3°

Determine the length of the belt required.

8. Design a cast iron trunk type piston for a single acting four stroke engine developing 75 kW per cylinder when running at 600 rpm. The other available data is as follows :

Maximum gas pressure	=	4.8 N/mm^2
Indicated mean effective pressure	=	0.65 N/mm^2
Mechanical efficiency	=	95%
Radius of crank	=	110mm
Fuel consumption	=	0.3 kg/BP/hour
Calorific value of fuel (higher)	=	$44 \times 10^3 \text{ kJ/kg}$
Difference of temperatures at the centre and edges of the piston head	=	200°C
Allowable stress for the material of the piston	=	33.5 MPa
Allowable stress for the material of the piston rings and gudgeon pin	=	80 MPa
Allowable bearing pressure on the piston barrel	=	0.4 N/mm^2
Allowable bearing pressure on the gudgeon pin	=	17 N/mm^2
